

Practice Questions for 2024-10-28-ApplsLinProg

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1 Questions

1. According to the production planning model presented, what does the variable x_i represent?
 - A. The demand for ice cream in month i
 - B. The surplus inventory at the end of month i
 - C. The production level in month i
 - D. The cost of changing production levels between consecutive months
 - E. The cost of storing surplus inventory
2. In the production planning model, what is the purpose of the constraint $x_i + s_{i-1} = d_i + s_i$?
 - A. To minimize the total production cost
 - B. To ensure that production plus beginning inventory equals demand plus ending inventory
 - C. To maximize the surplus inventory at the end of each month
 - D. To minimize the cost of changing production levels
 - E. To balance the average demand with the production level
3. What does the term $\sum_{i=1}^{12} 50|x_i - x_{i-1}|$ represent in the objective function of the production planning model?
 - A. The total cost of storing surplus inventory
 - B. The total demand for ice cream over 12 months
 - C. The total cost of changing production levels between consecutive months
 - D. The total production cost over 12 months
 - E. The average monthly demand

2 Answers

1. According to the production planning model presented, what does the variable x_i represent?
 - A. Answer A is incorrect because d_i represents the demand for ice cream in month i .
 - B. Answer B is incorrect because s_i represents the surplus inventory at the end of month i .
 - C. Answer C is correct because x_i is defined as the production at month i .
 - D. Answer D is incorrect because the cost of changing production levels is represented by $50|x_i - x_{i-1}|$.
 - E. Answer E is incorrect because the cost of storing surplus inventory is represented by $20s_i$.
2. In the production planning model, what is the purpose of the constraint $x_i + s_{i-1} = d_i + s_i$?
 - A. Answer A is incorrect because the objective function minimizes the total cost, not this constraint.
 - B. Answer B is correct because the constraint ensures that the production in month i plus the surplus from the previous month equals the demand in month i plus the surplus at the end of month i .
 - C. Answer C is incorrect because the constraint is about balancing inventory and demand, not maximizing surplus.
 - D. Answer D is incorrect because the cost of changing production levels is part of the objective function, not this constraint.
 - E. Answer E is incorrect because the constraint is about balancing inventory and demand, not the average demand.
3. What does the term $\sum_{i=1}^{12} 50|x_i - x_{i-1}|$ represent in the objective function of the production planning model?
 - A. Answer A is incorrect because the cost of storing surplus inventory is represented by $\sum_{i=1}^{12} 20s_i$.
 - B. Answer B is incorrect because the total demand is represented by $\sum_{i=1}^{12} d_i$, not this term.
 - C. Answer C is correct because the term represents the cost of changing production levels, where 50 is the cost per unit change.
 - D. Answer D is incorrect because the total production cost is not directly represented by this term alone.
 - E. Answer E is incorrect because the average monthly demand is not directly represented by this term.